Analytical / density-aware place

Analytical lite first pulls for wirelength, then spreads for density with pads A and D fixed

The idea

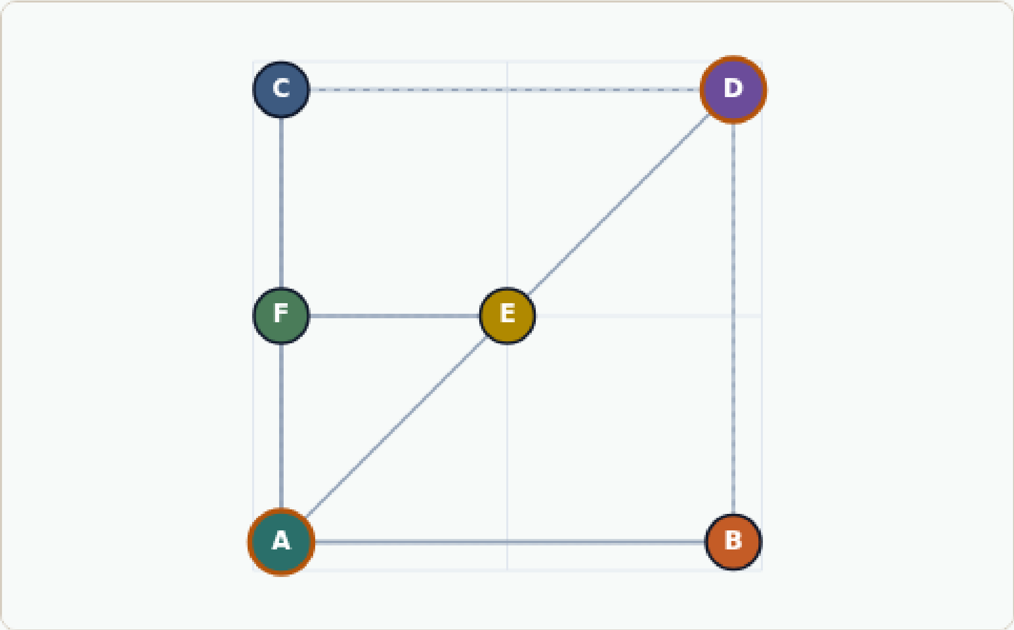
- Wirelength stage clusters
- Watch both HPWL and overflow
- <!-- algorithm-walkthrough -->

Wirelength stage clusters first

ANALYTICAL PLACE

Wirelength stage clusters first

Step 1 / 5 · wl-stage · module02-05-analytical-place



Explanation

Analytical lite starts like force/quadratic: pull for wirelength with pads A and D fixed. Clustering cuts HPWL but can overload bins.

- Force + quadratic wirelength stage
- Pads A, D fixed
- Clusters before spreading

```
Start HPWL: 52
Goal: cut WL without total collapse
```

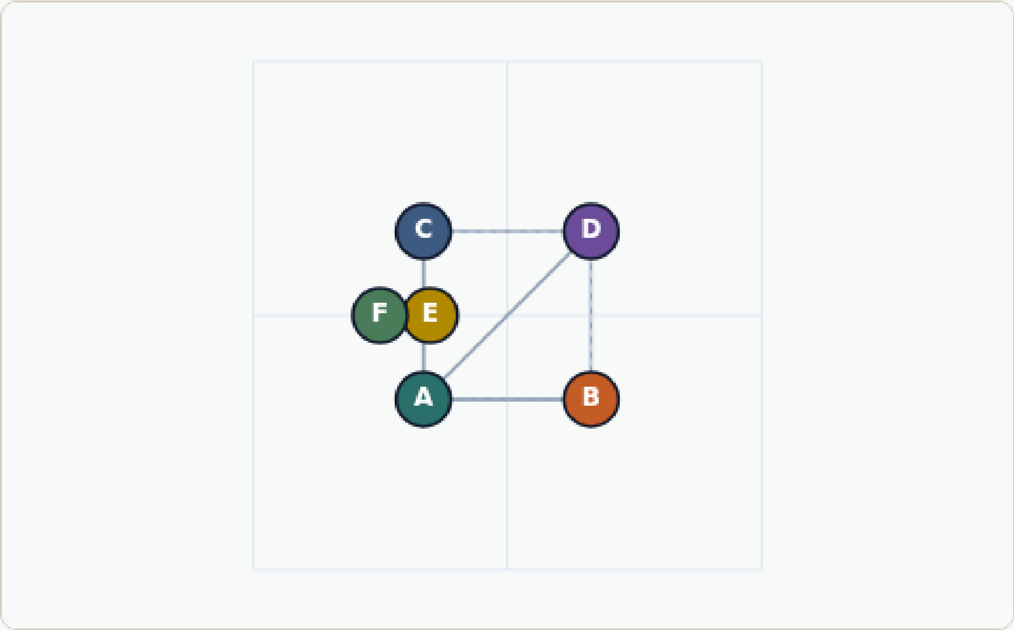
Wirelength stage clusters first

Density stage pushes overloaded bins

ANALYTICAL PLACE

Density stage pushes overloaded bins

Step 2 / 5 · density-stage · module02-05-analytical-place



Explanation

A density-repulsion stage pushes cells out of crowded two-by-two bins, then a light reconnect keeps HPWL from exploding.

- Count cells per bin
- Repel from crowded bin centers
- Light force reconnect afterward

```
2×2 density grid
Watch HPWL and overflow together
```

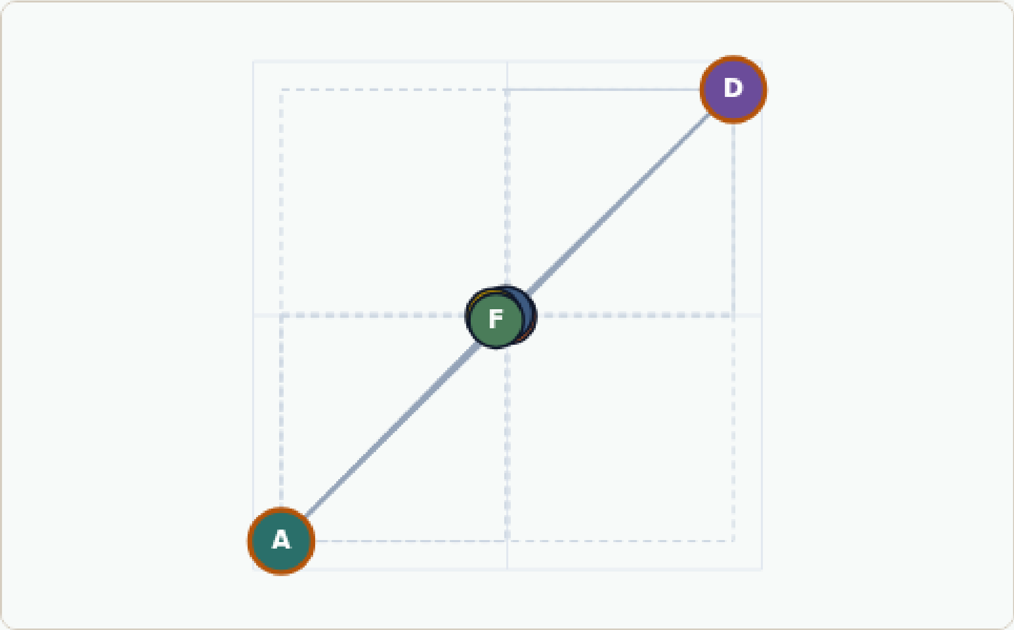
Density stage pushes overloaded bins

After analytical: about forty-eight point one

ANALYTICAL PLACE

After analytical: about forty-eight point one

Step 3 / 5 · after-anal · module02-05-analytical-place



Explanation

The combined solve lands near forty-eight point one—close to quadratic, deliberately above free force, because spreading fights pure collapse.

- Starter 52 → ≈48.1
- Near quadratic, above force
- Density-aware, not full legalizer

```
analyticalHpwlAfter: 48.1
Measured: 48.1
```

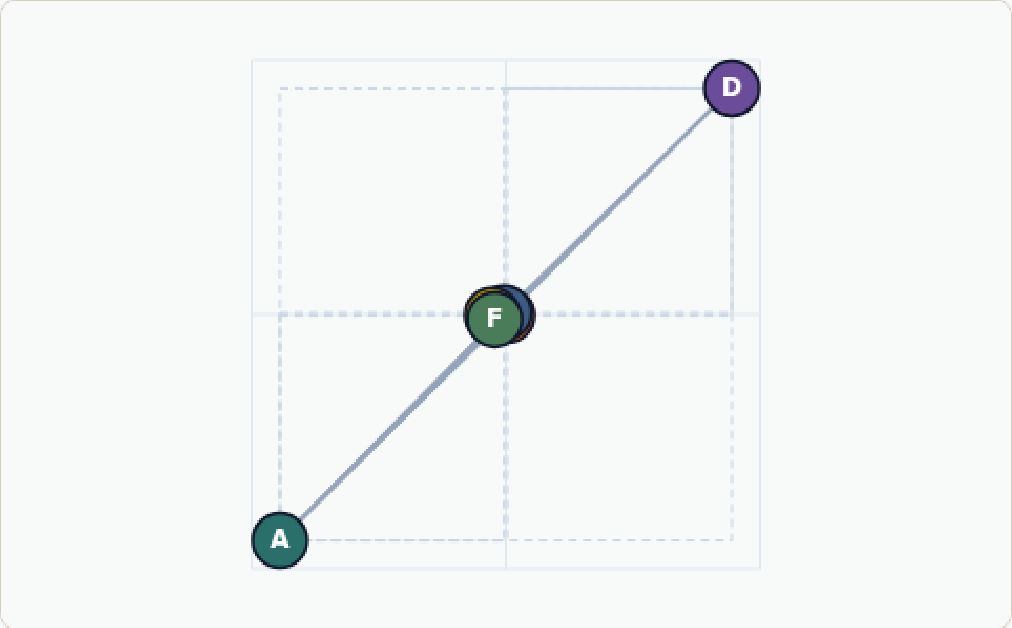
After analytical: about forty-eight point one

Report HPWL and overflow

ANALYTICAL PLACE

Report HPWL and overflow

Step 4 / 5 · both-metrics · module02-05-analytical-place



Explanation

Winning wirelength while overflowing every bin is not analytical success. Quote both metrics after the density stage.

- HPWL ≈ 48.1
- Bin overflow still matters
- Pads remain fixed

```
HPWL: 48.1
quadratic compare: 48
```

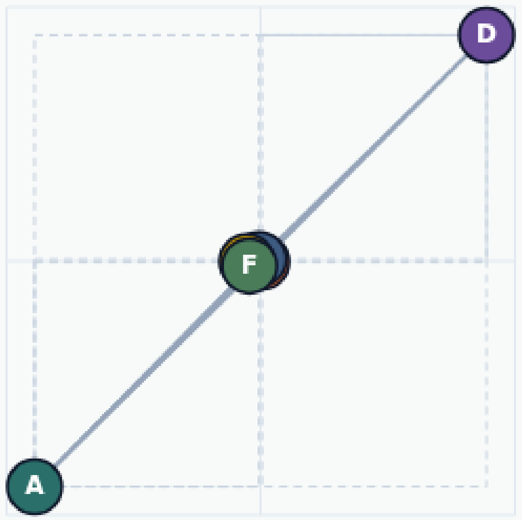
Report HPWL and overflow

Wirelength then density

ANALYTICAL PLACE

Wirelength then density

Step 5 / 5 · takeaway · module02-05-analytical-place



Explanation

Analytical lite is a two-act play: cluster for wirelength, spread for density. Lock pads and iteration knobs so forty-eight point one stays reproducible.

- WL stage → density stage
- ≈48.1 on this seed
- Next: simulated annealing

```
analyticalHpwlAfter: 48.1
```

Wirelength then density

Browser lab track

- In the browser lab track, open the **analytical-place** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track, open this module’s examples and the course `common/` solvers
- Parse `tiny_place.json`, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you’re ready, take the short quiz, then continue to the next module